

SENTRA

Safety Event Network for Threat Recognition & Anticipation

System architecture, test and gap report

Oil India Limited - Smart India Hackathon 2026, Problem Statement 26165 - Team WellDropp

322/322

automated tests passing

1.00

recall, 42 authored cases

2.8 ms

engine time per report

0

real Oil India reports seen

Version 2.0.0 - prepared 25 September 2026 - every number in this report was measured from the code on branch **tedo** at the time of writing, by the scripts listed in Appendix B.

What this report covers

Section	
1	Executive summary
2	How SENTRA actually works
3	System design
4	How the SIF precursor intelligence decides
5	Models - what is used now, and what fits better
6	Datasets - what was used, and what must be collected
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10	The recommended solution
A	Every test, by area
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1 Executive summary

SENTRA reads unsafe-act / unsafe-condition and near-miss reports and finds the ones carrying **serious-injury or fatality (SIF) potential** - reports where a high-energy source met a failed or missing critical barrier, whatever the outcome happened to be. It runs entirely on the operator's machine, explains every verdict, ranks recurring precursors by density, and never closes a report by itself: anything that matters goes to a person, whose decision becomes a training label.

What was done for this report. The whole codebase was reviewed module by module; every one of the 322 automated tests was run and recorded individually (all pass, 63 s); the engine was scored against both labelled sets in the repository; the learned model was trained and cross-validated; and the engine was timed. The findings are below, and the gaps are in Section 8.

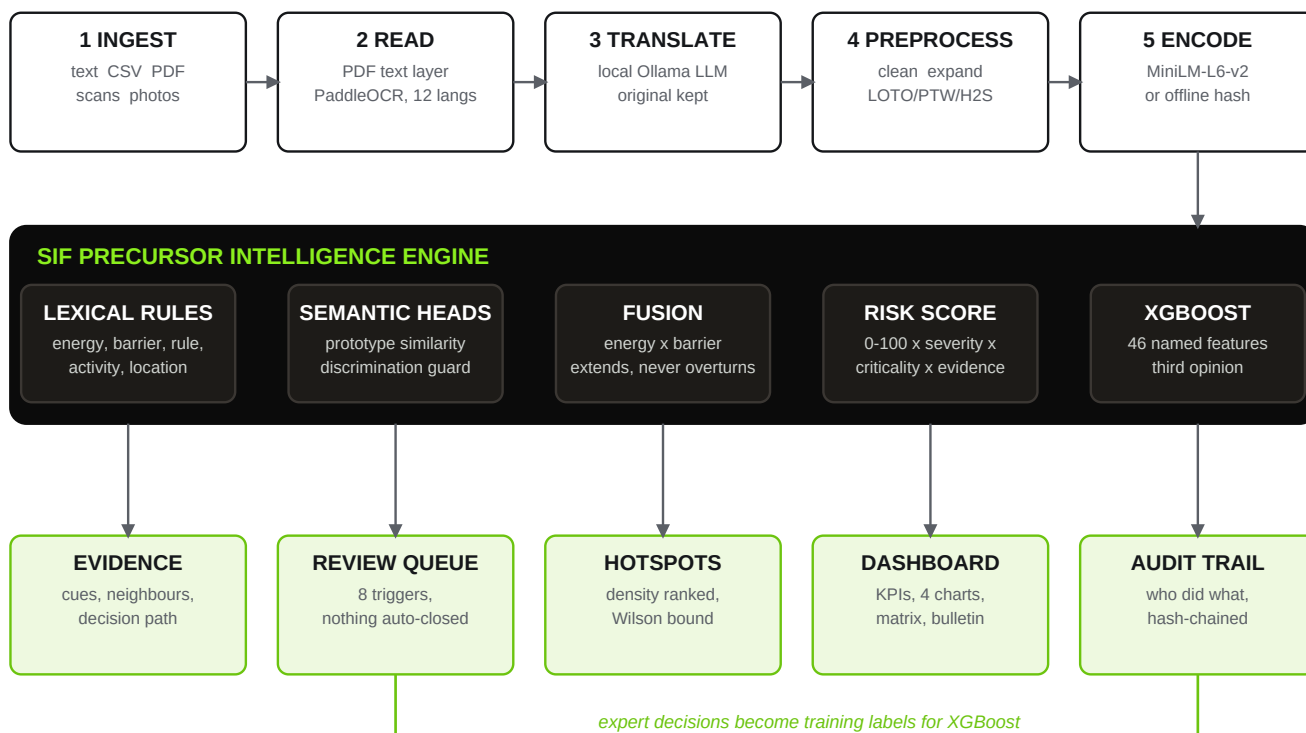
The single most important finding. Every accuracy figure in this report is perfect or nearly so - and that is a warning, not a result. Both labelled sets were written by the team, the engine's vocabulary was extended after seeing which cases it missed, and the learned model's strongest feature is the rule engine's own score. These numbers show that the system is **internally consistent**. They say nothing yet about how it performs on Oil India's real reports, because it has never seen one. Section 6 specifies the data that would change that; it is the highest-value next step by a wide margin.

Headline findings

Finding	
Works	All 322 tests pass: every pipeline stage, the review bench, OCR and LLM handling, every page of the interface, the release pipeline, and the new sign-in and audit chain.
Works	The engine is fast - 2.8 ms per report, about 21,739 reports a minute on one core - so the interface, not the analysis, is the only speed constraint.
Works	No confirmed SIF finding is ever closed without a person: 0 of 42 on the evaluation set reach nobody, by construction and by test.
Gap	No real data. All 127 labelled or sample reports in the repository are authored.
Gap	The learned model is a distillation of the rules until real reviewed labels exist.
Gap	Word documents (.docx) are not read; there is no asset/equipment identity; there is no work-hold output and no central server.
Added	Sign-in with roles, every report and decision attributed to a named person, and a hash-chained audit trail that detects tampering (Section 9).

2 How SENTRA actually works

A report travels five stages to reach the engine, and the engine's verdict travels to five places. Nothing in the path needs a network connection: the encoder, the OCR models and the optional language model are all local.



Stage	Module	What it does
1 Ingest	sif/ocr.py	Accepts pasted text, CSV exports, PDFs, scanned pages and photographs (.txt .md .log .csv .tsv .pdf and seven image formats).
2 Read	sif/ocr.py	A PDF's text layer is read exactly with pypdfium2; a page without one, or an image, goes through PaddleOCR in one of 12 languages. Every line keeps its OCR confidence.
3 Translate	sif/llm.py	A non-Latin report is translated to English by a local Ollama model before analysis. The original wording is kept as the record; the English is what the engines read.
4 Preprocess	sif/preprocessing.py	Cleans the text, expands field shorthand (LOTO, PTW, GGS, H2S) for the encoder while keeping the reporter's own words for the patterns, and splits sentences.
5 Encode	sif/encoders.py	all-MiniLM-L6-v2 sentence embeddings (384-d). If that model cannot load, a deterministic hashing encoder takes over - and is barred from raising a flag of its own.
Engine	sif/heads.py, scoring.py	Decides SIF potential, the IOGP rule, activity, location, energy source and failed barrier, and a 0-100 risk score (Section 4).
Outputs	evidence, review, patterns	Evidence for every field, a queue for human review, density-ranked hotspots, the dashboard and a plain-English bulletin.
Learn	sif/mlops.py	Reviewed decisions train an XGBoost model on 46 named features; each run is logged to MLflow; the model becomes a third opinion.

3 System design

Four layers, each depending only on the one below. The analysis layer imports nothing from Qt, so it can be run from a script, a test or a future server without the interface.

PRESENTATION

app.py (black/lime) - app2.py (deep navy) - ui/ skins - ui2/ pages - sign-in dialog

CONTROLLER

main2.MainWindow - role guard on every action - 4 worker threads (analysis, extraction, training, probes)

ANALYSIS

sif/ pipeline - lexical - encoders - heads - evidence - scoring - patterns - review - mlops - llm - ocr

STORAGE

%APPDATA%\SIF Insight Console - users.json - audit.jsonl (chained) - review decisions - logs - models - mlflow.db

Two entry points, one console

app.py and **app2.py** build the same window from the same controller (main2.py) and differ only by a palette and a style sheet. A fix to any capability lands in both. A skin is applied in two halves, because Qt style sheets cannot reach widgets that paint themselves: the palette is set *before* the window is built, the style sheet after. Letter case and rail icons, which no style sheet can change, are switches in ui.theme.LOOK that every skin sets in full.

Concurrency

Worker thread	Runs	Why it is off the interface thread
AnalysisWorker	Encoder load and every pipeline stage; streams one row at a time	A 90 MB model load or a thousand-report import must not freeze the window.
ExtractionWorker	PDF reading and OCR, including first-run model download	PaddleOCR's first run downloads models.
TrainingWorker	XGBoost training and the MLflow run	Cross-validation takes seconds.
Probe workers	OCR and LLM availability checks, the update check	A socket to a stopped Ollama can block for seconds.

Where the data lives

File	Format	Written by	Notes
users.json	JSON	Sign-in, account management	Salted PBKDF2 digests only; never a password. Atomic write.
audit.jsonl	JSON lines	Every action	Append-only, fsync'd, hash-chained. Separate from the debug log, which rotates.
review_decisions.json	JSON	The review bench	Append-only in meaning: a changed mind appends a new entry.
settings.json	JSON	Preferences	Reviewer, rail width, update checks.
logs/	rotating text	logging	Diagnostics; 1 MB x 3.
models/, mlflow.db	JSON, SQLite	Training	The XGBoost booster and every run's metrics.

All of it sits in %APPDATA%\SIF Insight Console. An installed build never writes into Program Files, which a standard user cannot write to (sif/paths.py).

4 How the SIF precursor intelligence decides

The whole engine rests on one definition, taken from the energy-barrier model that has replaced the Heinrich triangle in modern SIF practice:

A report has SIF potential when a high-energy source was present AND a critical barrier failed, was missing, bypassed or was not verified.

Expressed continuously: $P(\text{SIF}) = \text{energy score} \times \text{barrier score}$

The product is the point. A report full of alarming words about a dropped glove scores nothing, because there is no high energy; a report about a live 11 kV feeder where isolation held scores nothing either, because no barrier failed. Severity wording alone can never create a flag, and a dismissive tone ("nothing serious to report") can never remove one.

Three readings, fused

Layer	How it reads a report	Its authority
Lexical rules (sif/lexical.py)	Curated patterns for 11 IOGP Life-Saving Rules, high-energy sources, critical barriers - including implicit failures such as "clipped to the handrail instead of the anchor point" - and site and activity vocabulary.	The backbone. What it flags stays flagged.
Semantic heads (sif/heads.py)	Cosine similarity between the report's embedding and prototype descriptions of each energy, barrier and rule.	Can add a flag the rules missed; can never remove one. A discrimination guard ignores it when every prototype scores alike.
XGBoost (sif/mlops.py)	46 named features - both SIF factors, energy and barrier families, severity weights, the rule, text statistics.	A third opinion. Where it disagrees with the pipeline, the report goes to a person.
Local LLM (optional)	Reads the narrative and answers the same question.	A fourth opinion. Never overrides; a disagreement is queued.

Risk score and review

Risk = 100 x P(SIF) x energy severity x barrier criticality x evidence factor, banded Critical (70+), High (50+), Medium (30+) and Low. It is ordinal - for ranking a queue - not an actuarial probability. A report is queued for a person on any of eight triggers, in this priority: rule/semantic disagreement, model disagreement, LLM disagreement, critical risk, thin evidence, unclassified exposure, energy with no barrier found, and - as a catch-all - any other confirmed finding. **No report the engine calls SIF-potential is ever closed without a person.**

Hotspots rank a cluster by the share of its reports that carry SIF potential, discounted by the Wilson lower bound so a 2-of-2 cluster cannot outrank a 12-of-20 one.

A real report, traced through the engine

Report NM-2601 from the sample corpus, exactly as the engine produced it with the offline encoder:

During preventive maintenance of the booster pump at OCS-4 the 11 kV feeder cable was left ungrounded and the breaker was not racked out. The technician had already started cable jointing when the permit was checked; no LOTO was applied and no isolation certificate had been raised.

Step	What the engine found
Energy cues	11 kV, breaker, cable jointing, feeder -> Electrical energy , high energy
Barrier cues	left ungrounded, no isolation certificate, no LOTO -> Energy isolation / LOTO not applied; Permit to work not followed
Rule cues	11 kV, breaker, feeder, isolation certificate -> Energy Isolation (confidence 0.89; next best Work Authorisation 0.18)
Activity, location	Equipment maintenance at Oil Collecting Station (OCS) - both from the lexical layer
P(SIF)	1.00 = energy 1.00 x barrier 1.00
Risk	100.0, Critical = P(SIF) 1.00 x energy 1.00 x barrier 1.00 x evidence 1.00
Semantic layer	Inactive: the offline encoder placed this report nearest "Toxic / Asphyxiant atmosphere" (0.36) - wrong - which is exactly why that encoder may rank but never flag. Decision path: <i>lexical rules only</i> .
Review	Queued - trigger Critical risk : "risk 100/100 - verify before it drives an intervention"
Plain English	"Electrical energy was uncontrolled because energy isolation / LOTO not applied or verified; permit to work absent, expired or not followed during equipment maintenance."

5 Models - what is used now, and what fits better

The present design is right for a system with **no labelled data**: a rule engine that needs none, an off-the-shelf encoder that needs none, and a small gradient-boosted model that can be trained on the first few hundred reviewed reports. What changes the answer is real data - and Indian-language reports.

Job	Used now	Better suited, to evaluate next	Why
Decide SIF potential	Lexical rules + P(SIF) fusion	Keep, and add a fine-tuned classifier beside it (below)	The rules are auditable and need no data; they should stay the floor that recall cannot fall below.
Sentence embedding	all-MiniLM-L6-v2, 384-d, English only	multilingual-e5-small / base, or para phrase-multilingual-MiniLM-L12-v2; BGE-M3 if the machine can afford it	Embeds Hindi, Tamil, Telugu and the rest directly, so meaning survives without a translation step that can lose it.
Learn from reviews	XGBoost on 46 engineered features (220 trees, depth 4)	SetFit (contrastive fine-tuning of a sentence encoder) once there are 8-64 labelled reports per class; later a fine-tuned DeBERTa-v3-small or MuRIL at 2,000+	SetFit is built for exactly this few-label regime; MuRIL and IndicBERT are pretrained on Indian languages.
Probability quality	Raw XGBoost probability	Isotonic or Platt calibration on a held-out set	A risk score that says 0.8 should be right about 80% of the time before anyone acts on it.
Translation	Local Ollama (llama3.2 3B default; gemma2 9B optional)	IndicTrans2 (AI4Bharat), run locally	Purpose-built for 22 Indian languages; a general 3B chat model is not a translator. gemma2 9B does not fit a typical plant GPU (measured 20%/80% CPU/GPU split).
Extract equipment	None - no asset field exists	GLiNER (zero-shot NER), or a small NER fine-tuned on tagged reports	Enables asset-level memory: repeat failures on the same pump, crane or well.
OCR	PaddleOCR, 12 languages	Keep; add Bengali, Gujarati, Punjabi, Malayalam, Odia, Assamese as recognisers mature	These six Indian scripts have no recogniser in the current build.

Recommendation. Keep the rule engine as the auditable floor. Replace the English encoder with a multilingual one. When 300-500 reports have been reviewed, train SetFit beside XGBoost and let both be opinions, not deciders - and only promote a model whose recall on real, held-out Oil India reports is at least the rule engine's. These are candidates to evaluate on real data, not results.

6 Datasets - what was used, and what must be collected

What is in the repository today

Set	Size	Labels	Used for	Real?
evaluation/labelled_reports.csv	42	25 SIF, 17 controls; IOGP rule	Scoring the engine	Authored
samples/training_corpus.csv	56	33 SIF, 23 controls; IOGP rule	Training XGBoost	Authored
samples/near_miss_reports.csv	18	none	End-to-end import tests; held out from tuning	Authored
sample_reports.csv	6	none	Batch-import demo	Authored
Seed incidents (sif/lexical.py)	5	none	One-click demo	Authored
samples/languages/, multilingual	6 + 5 blocks	none	Hindi, Marathi, Tamil, Telugu, Kannada, Urdu ingestion	Authored
permit PDF, scanned PNG, shift log	3 files	none	PDF text layer, OCR and text-block paths	Authored

The labelled sets are carefully built - every positive has a twin in which the barrier *held*, and minimising wording appears on both sides - but they were written by the same team that wrote the vocabulary. They test that the logic is coherent. They cannot test whether it matches how Oil India's crews actually write.

What must be collected

Specification	
Source	Oil India's own UA/UC and near-miss records - the HSE reporting system's export - including free-text narratives as written, in every language they arrive in; plus incident investigation reports where the outcome is known.
Volume	300-500 reviewed reports to train SetFit and trust XGBoost; 2,000+ to fine-tune a transformer; at least 50 true SIF precursors in any evaluation set, so recall is measured on more than a handful.
Labels per report	SIF potential (yes / no / unclear); high-energy source; failed barrier; IOGP rule; activity; location; asset or equipment tag ; actual and potential severity; language.
Labelling protocol	Two trained HSE reviewers label independently; disagreements are adjudicated; Cohen's kappa is reported and should be 0.7 or higher before the labels are trusted. SENTRA's review bench already records this per decision.
Splits	By time, not at random: train on older reports, evaluate on the most recent quarter - which is how the system will be used.
Balance	SIF precursors are rare. Keep the natural rate in evaluation; oversample only in training.
Privacy	Remove names and employee IDs before labelling; keep sites and equipment, which carry the signal.
Supplementary	OSHA Severe Injury Reports and fatality investigation summaries (public, US) are real narratives with known outcomes - useful for pretraining the classifier, never for evaluating it on Oil India's operations.

7 Testing - method and results

Method. The suite is Python unittest, run headless (offscreen Qt platform) with the deterministic offline encoder, so every run is reproducible with no network. It exercises every layer: unit tests for each engine stage; end-to-end tests that drive the real window against the files in samples/ from import through review to a trained model; and some that measure rendered pixels, because a few interface faults - a widget painting over the rail, a checkbox that draws nothing, a table wider than its page - are invisible to any other kind of test. Every test's outcome and time was recorded individually; Appendix A lists all of them.

Functional area	Test classes	Tests	Passed	Time (s)
Pipeline and engine	11	63	63	3.6
Human review	5	46	46	29.4
Learning (MLOps)	4	20	20	2.1
Ingestion and OCR	6	37	37	8.9
Local LLM and translation	3	23	23	0.8
Interface	10	62	62	11.9
Audit and logging	3	18	18	0.0
Sign-in and roles	2	25	25	4.3
Release and install	4	28	28	0.7
Total	48	322	322	63.1

Engine accuracy

Measure	Evaluation set (42)	Training corpus (56)
Recall - SIF precursors found	1.000 (25 of 25)	1.000 (33 of 33)
Precision - flags that were real	1.000 (0 false)	1.000 (0 false)
IOPG rule right, on true positives	0.960	0.879
Queued for a person	42 of 42	56 of 56

Learned model (XGBoost, cross-validated on the 56-report corpus): accuracy 1.00, precision 1.00, recall 1.00, ROC AUC 1.00. Top features: p_sif (0.76), barrier_criticality (0.14), lexical_flag (0.09). Small corpus (56 reports, 23 in the minority class): regularisation relaxed so the trees can split. Metrics from this few samples are indicative only.

Reading these honestly. Perfect scores on authored data are a ceiling, not a forecast. The engine's vocabulary was extended after seeing its misses on the 42-case set, which is fitting to the test. The model's top feature, p_sif at 0.76, is the rule engine's own output - it has learned to agree with the rules, which is distillation, not new knowledge. The first measurement that means something will be on real Oil India reports labelled by Oil India reviewers.

Speed

The engine analyses a report in **2.76 ms** with the offline encoder (about 21,739 a minute). Earlier measurement found the interface, not the engine, was the bottleneck: it spent ~94 ms per report on a sleep, a refresh every five rows and a per-row scroll; that is now 4-6 ms. The sentence-transformer encoder adds roughly an order of magnitude per report and remains fast enough for any realistic import.

8 Software gaps

Ordered by what each costs if left as it is. **Fixed** marks what this update closed.

Severity	Gap	What it costs	The fix
High	No real data - every report and label is authored	Accuracy on Oil India's reports is unknown	Collect and label per Section 6; measure before any operational use
High	The model is trained on the pipeline's own verdicts until reviews accumulate	It agrees with the rules and adds nothing	Train only on reviewed labels; SetFit at 8-64 per class
High	One machine, one database - no central server	Attribution and decisions are per workstation; two sites cannot share a hotspot	A small central service (Section 10, P3)
Fixed	No sign-in; the reviewer was a free-text box	Anyone could decide under any name	Accounts, roles, attributed decisions (Section 9)
Fixed	Audit trail could be edited without trace	"Who did what" could not be proven	Hash-chained entries with on-screen verification
Medium	Word documents (.docx) not read	A common report format is rejected at ingest	python-docx reader, about twenty lines
Medium	No asset or equipment identity	Repeat failures on the same equipment are invisible	Equipment NER and a fifth hotspot grouping
Medium	No work-hold output	Detection stops at the queue; nothing says "stop the job"	A threshold on high energy + failed critical barrier + critical risk, surfaced on the report and queue
Medium	English-only encoder; translation by a general chat model	Meaning can be lost before analysis	Multilingual encoder; IndicTrans2
Medium	Probabilities not calibrated	A 0.8 may not mean 80%	Isotonic calibration on held-out reviews
Medium	Six Indian scripts have no OCR recogniser	Bengali, Gujarati, Punjabi, Malayalam, Odia, Assamese scans cannot be read	Named honestly in the interface; revisit as PaddleOCR adds them
Low	Decision log rewrites the whole file per decision	25 ms at 3,000 decisions; grows linearly	Append-only JSON-lines, as the audit trail already is
Low	Installer is not code-signed; 1.64 GB	SmartScreen warning; close to GitHub's 2 GB asset limit	Organisation certificate; ship the slim build plus an engine pack
Low	Truncating the end of the audit trail leaves a valid shorter chain	Removed final entries are undetectable without the noted head	Periodic head export to a second location

9 This update: sign-in, roles and a record of who did what

The console knew only the operating-system user - shared on a plant workstation - and a reviewer name typed into a box. Neither could answer the question an auditor asks: **which person analysed this report, and which person decided it?** Every session now begins with a sign-in, and everything recorded afterwards carries the account that did it.

Left: the first start on a machine. There is no default password - with no accounts, the page creates the administrator. Right: a refused sign-in, which is recorded with the name that was tried.

Roles

Role	Can	Cannot
Viewer	Read every page	Ingest, analyse, decide, train, clear, configure, manage accounts
HSE Analyst	+ ingest and analyse reports, export results	Decide review cases, train, clear
HSE Expert	+ decide review cases, train the model on them	Clear records, configure engines, manage accounts
Administrator	+ clear records, configure engines, manage accounts, export the audit trail	Demote or disable the last administrator

The check sits on the controller action, not the button, so it holds for clicks, keyboard shortcuts, menus and the workflow map alike. A refusal is shown and recorded.

How accounts are protected

Passwords	Never stored. Each account keeps a random 16-byte salt and a PBKDF2-HMAC-SHA256 digest at 600,000 iterations (OWASP's 2023 figure), compared in constant time.
Guessing	Five wrong passwords lock the account for five minutes. A wrong password and an unknown username give the same message in the same time, so neither reveals which usernames exist.
New accounts	An administrator never chooses someone's password: the console issues a one-time password that must be replaced at first sign-in.
Lock-out safety	The last active administrator cannot be demoted or disabled.

File Help

SENTRA

WORKFLOW MAP

INGEST AND OCR

DASHBOARD

REPORTS AND EVIDENCE

RISK HOTSPOTS

HUMAN REVIEW

ANALYTICS

ENGINES

ACTIVITY

SETTINGS

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encoder: pending · LLM: off · model: none

D. Manikandan
Administrator

Activity and access

Who signed in, what each person did, and whether the record has been altered since.

People

USER	NAME	ROLE	LAST SIGN-IN	SIGN-INS	REPORTS ANALYSED	DECISIONS	STATUS
a.baruah	A. Baruah	HSE Expert	2026-09-25 ...	1	9	3	active
d.manikandan	D. Manikandan	Administrator	2026-09-25 ...	1	0	0	active
k.das	K. Das	Viewer	-	0	0	0	active
r.sharma	R. Sharma	HSE Analyst	2026-09-25 ...	1	9	0	active

Add an account

Issue a new password

Change role

Disable / enable

Activity

Show Everyone

TIME	USER	ROLE	ACTION	DETAIL
2026-09-25 ...	d.manikandan	admin	signed in	username: d.manikandan · role: admin
2026-09-25 ...	d.manikandan	admin	console started	build: 2 · version: 2.0.0
2026-09-25 ...	a.baruah	reviewer	console closed	reports: 9 · outstanding: 6
2026-09-25 ...	a.baruah	reviewer	signed out	username: a.baruah
2026-09-25 ...	a.baruah	reviewer	review decision	reference: NM-2617 · decision: confirmed · trigger: Critical risk · overturns_engine: False
2026-09-25 ...	a.baruah	reviewer	review decision	reference: NM-2615 · decision: confirmed · trigger: Critical risk · overturns_engine: False
2026-09-25 ...	a.baruah	reviewer	review decision	reference: NM-2610 · decision: confirmed · trigger: Thin evidence · overturns_engine: True

Integrity of the record

Chain intact - 16 entr(ies) verified - head b82a46b28ce9

Head of the chain: b82a46b28ce9765051c51686a854f07927ecf1613acad0fe9de449f9c56a51c1

Verify the trail

SAFETY FIRST

No report is closed by the engine.

Every flag carries its evidence.

Oil India Limited · PS 26165 · prototype output, not a statutory record

Ready. Start on the workflow map.

The Activity page: each person's sign-ins, reports analysed and decisions, read from the trail; the trail filterable to one person; account management for administrators; and the state of the chain.

What is recorded, and under whom

Event	Recorded with
Signed in / signed out / sign-in refused / account locked	The account (or, when refused, the name that was tried), role, session id
Reports analysed	The account, the count and the reference of every report in the batch
Each analysed report	"Analysed by" and the time, shown in its evidence panel and exported with it
Review decision	Recorded under the signed-in person; the reviewer box can no longer be edited
Permission refused	The account, what it tried, and which permission it lacked
Account created, role changed, password reset, disabled	The administrator who did it

A trail that can prove it has not been altered

Each audit entry carries the SHA-256 of the entry before it and of itself. Editing, inserting or deleting a line breaks the chain at exactly that line, and the Activity page says where. Removing lines from the *end* leaves a shorter chain that still verifies - so the head hash is shown on screen, for an auditor to note and compare later.

What this is not. A security boundary against someone with access to the files. The accounts and the trail live on the operator's own disk, and anyone who can edit them can edit them. What sign-in buys is **attribution**: an honest operator cannot act under someone else's name by accident, and a dishonest one leaves a trail whose tampering can be detected. Tested by 32 new tests, all passing.

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12

10 The recommended solution

The intelligent path is not a bigger model. It is **the right data, a human in the loop who is named, and models that are only promoted when they beat the rules on real reports**. In order:

Phase	What	Done when
P0 Data (weeks 1-8)	Pilot at one asset. Import the real UA/UC export. Two HSE reviewers label through SENTRA's own review bench - which now records who decided what - with adjudication of disagreements.	500 reviewed reports; Cohen's kappa of 0.7 or more; the first honest recall figure on real reports.
P1 Models (weeks 6-12)	Multilingual encoder in place of the English one. IndicTrans2 for translation. SetFit trained on reviewed labels beside XGBoost. Isotonic calibration.	A model is promoted only if its recall on the latest held-out quarter is at least the rule engine's, at no worse precision.
P2 Capabilities (weeks 8-14)	Work-hold recommendation on high energy + failed critical barrier + critical risk. Equipment extraction and asset-level hotspots. Word documents at ingest.	Each has tests and appears on the report, the queue and the hotspots page.
P3 Platform (weeks 12-20)	A small central service holding accounts, decisions and the audit chain, so attribution and hotspots span sites; workstations sync when connected and work offline between. Signed installer; slim build plus an engine pack.	Two workstations at two sites see one queue, one trail and one set of hotspots.

Three rules that should not change

1	The rules are the floor. A learned model may add a flag; it may not remove one the rules raised. Recall can only grow.
2	Nothing is closed by software. Every confirmed finding reaches a named person.
3	Every number is measured. No accuracy figure is quoted unless it was measured on real, held-out reports, by a script anyone can run.

Appendix A Every test, by area

All 322 tests, each run and recorded individually. P = passed. Times in seconds.

Pipeline and engine (63)

Test	Result	s
Pipeline analyze many skips blanks	P	0.00
Pipeline blank report is handled	P	0.00
Pipeline every result reports its encoder	P	0.00
Pipeline evidence carries the audit trail	P	0.00
Pipeline non sif report scores zero risk	P	0.00
Pipeline offline backend raises no semantic flag	P	0.00
Pipeline required fields survive the rewrite	P	0.00
Pipeline risk score is bounded and banded	P	0.00
Pipeline seed corpus matches the lexical baseline	P	0.00
LexicalEngine blank input returns fallbacks	P	0.00
LexicalEngine electrical isolation failure is sif	P	0.00
LexicalEngine height with failed fall protection is sif	P	0.00
LexicalEngine high energy with controls in place is not sif	P	0.00
LexicalEngine low energy observation is not sif	P	0.00
LexicalEngine returns all required keys	P	0.00
LexicalEngine rule scores are exposed for fusion	P	0.00
Heads calibration is bounded and capped for the fallback	P	0.00
Heads margin detects a flat ranking	P	0.00
Heads semantic index ranks every label	P	0.01
Scoring band thresholds	P	0.00
Scoring housekeeping barrier ranks below isolation	P	0.00
Encoders auto backend always returns a working encoder	P	0.00
Encoders explicit hashing backend never downloads	P	0.00
Encoders fallback encoder is marked non semantic	P	0.00
Encoders similarity is bounded and non negative	P	0.00
Encoders transformer encoder defers loading	P	0.00
Encoders unavailable transformer raises when demanded	P	3.46
Encoders vectors are unit length and deterministic	P	0.00
Preprocessor blank input yields empty document	P	0.00
Preprocessor expands domain abbreviations for the encoder	P	0.00
Preprocessor segments sentences without splitting on no	P	0.00
Preprocessor tokens are lowercased	P	0.00
EngineQuality every confirmed finding reaches a person	P	0.00
EngineQuality no precursor is both missed and unqueued	P	0.00
EngineQuality precision stays above the floor	P	0.00
EngineQuality recall stays above the floor	P	0.00
EngineQuality the labelled set is balanced enough to mean something	P	0.00
EngineQuality the negative controls do not flag	P	0.04
Intelligence activities are ranked as hotspots	P	0.01
Intelligence critical reports are queued for review	P	0.00
Intelligence hotspots rank by evidence adjusted density	P	0.00

Test	Result	s
Intelligence kpis	P	0.00
Intelligence offline mode never reports disagreement	P	0.00
Intelligence repeat location becomes a hotspot	P	0.00
Intelligence singleton groups are not hotspots	P	0.00
Intelligence thin evidence is flagged	P	0.00
NarrativeGeneration a clear queue says so explicitly	P	0.00
NarrativeGeneration a flagged report names the rule and the risk	P	0.00
NarrativeGeneration an empty report is named not hallucinated	P	0.00
NarrativeGeneration an unflagged report does not claim fatal potential	P	0.00
NarrativeGeneration every cited reference is a real analysed report	P	0.00
NarrativeGeneration language coverage appears only for translated reports	P	0.00
NarrativeGeneration minimizing language is named only when present	P	0.00
NarrativeGeneration sections are silent when there is nothing to say	P	0.00
NarrativeGeneration the bulletin headline matches the kpis it was built from	P	0.00
NarrativeGeneration translation is disclosed when the report was translated	P	0.00
MinimizingLanguage dismissive wording alone does not manufacture a finding	P	0.01
MinimizingLanguage dismissive wording does not suppress a real finding	P	0.01
MinimizingLanguage the note is silent when there is nothing to flag	P	0.00
MinimizingLanguage the review reason names the mismatch	P	0.00
EnergyWithoutBarrierTrigger a flagged report is not queued twice for it	P	0.00
EnergyWithoutBarrierTrigger an ordinary report still reaches nobody	P	0.00
EnergyWithoutBarrierTrigger high energy with no barrier reaches a person	P	0.00

Human review (46)

Test	Result	s
ReviewBench a decision leaves the queue and becomes a label	P	0.12
ReviewBench a report with no fatal potential does not claim one	P	0.12
ReviewBench a report with no reference is given one	P	0.13
ReviewBench clearing the corpus forgets the reports not the decisions	P	0.12
ReviewBench decided reports come back when asked for	P	0.13
ReviewBench deciding never skips the next report	P	0.16
ReviewBench every case opens with a plain english brief	P	0.15
ReviewBench selecting a row shows the whole case	P	0.18
ReviewBench the audit trail records what the operator did	P	0.23
ReviewBench the brief omits fields the report never stated	P	0.22
ReviewBench the header counts outstanding work not decided work	P	0.23
ReviewBench the queue shows what is outstanding	P	0.22
ReviewBench the same report analysed twice is one report	P	0.22
ReviewBench the trail keeps every entry including the changed ones	P	0.24
ReviewBench too small a corpus is refused rather than trained	P	0.22
ReviewBench training prefers human labels once there are enough	P	0.23
ReviewBench undo puts the report back in the queue	P	0.22
ReviewDecisions a corrupt file reads as empty	P	0.00
ReviewDecisions a decision reaches disk immediately	P	0.00
ReviewDecisions a read only location is reported not raised	P	0.00

Test	Result	s
ReviewDecisions an unknown decision is refused	P	0.00
ReviewDecisions an unreviewed report is never guessed at	P	0.00
ReviewDecisions changing a decision supersedes without erasing	P	0.00
ReviewDecisions decided reports leave the queue	P	0.00
ReviewDecisions export carries the whole trail	P	0.00
ReviewDecisions fingerprint survives reformatting but separates reports	P	0.00
ReviewDecisions overturning the engine is recorded as such	P	0.00
ReviewDecisions unclear is recorded but never becomes a label	P	0.00
ReviewDecisions undo withdraws only the last entry	P	0.00
ConfirmedFindingsAlwaysReachAPerson a high band confirmed finding is queued	P	0.00
ConfirmedFindingsAlwaysReachAPerson a medium band confirmed finding is queued	P	0.00
ConfirmedFindingsAlwaysReachAPerson critical band still wins on specificity	P	0.00
ConfirmedFindingsAlwaysReachAPerson no sif potential result is ever left unqueued	P	0.00
DecisionTrailScaleAndClearing an unchanged trail is not rebuilt	P	2.33
DecisionTrailScaleAndClearing clearing an empty queue says so rather than asking	P	0.25
DecisionTrailScaleAndClearing clearing an empty trail says so rather than asking	P	0.22
DecisionTrailScaleAndClearing clearing the trail asks first and says what it costs	P	0.23
DecisionTrailScaleAndClearing confirming clears the trail and records that it happened	P	0.29
DecisionTrailScaleAndClearing confirming empties the queue but keeps the decisions	P	0.39
DecisionTrailScaleAndClearing refreshing stays cheap as the trail grows	P	21.04
DecisionTrailScaleAndClearing the clear button is on the trail tab of both builds	P	0.62
DecisionTrailScaleAndClearing the clear button is wired to the confirmation	P	0.26
DecisionTrailScaleAndClearing the queue has a clear of its own that asks first	P	0.22
ReviewToLabelsEndToEnd the model trains on the decisions a person made	P	0.25
ReviewToLabelsEndToEnd the trail exports for an auditor	P	0.06
ReviewToLabelsEndToEnd working the queue empties it and leaves a trail	P	0.08

Learning (MLOps) (20)

Test	Result	s
MLOps feature vector matches the named columns	P	0.00
MLOps features are empty safe	P	0.00
MLOps mlflow run is logged and listed	P	1.02
MLOps small corpus relaxes regularisation	P	0.00
MLOps train predict save load roundtrip	P	0.92
MLOps training requires both classes and enough rows	P	0.00
MLOps untrained service predicts none	P	0.00
ModelInPipeline attached model scores every row	P	0.04
ModelInPipeline model disagreement reaches the review queue	P	0.01
ModelInPipeline provider returning none is ignored	P	0.01
TrainerCLI dry run succeeds and bad input fails	P	0.04
TrainerCLI label column is auto detected	P	0.00
TrainerCLI missing narrative column is rejected	P	0.00
TrainerCLI reads narratives references and labels	P	0.00
TrainerCLI without labels none is returned	P	0.00
TrainingCorpus every labelled precursor is found	P	0.00

Test	Result	s
TrainingCorpus it does not overlap the evaluation set	P	0.00
TrainingCorpus it is usable as training data	P	0.00
TrainingCorpus no negative control flags	P	0.00
TrainingCorpus the three phrasings this set exposed stay readable	P	0.00

Ingestion and OCR (37)

Test	Result	s
DocumentExtraction batch records failures instead of raising	P	0.00
DocumentExtraction probe returns a verdict either way	P	0.00
DocumentExtraction reads pdf text layer without ocr	P	0.01
DocumentExtraction reads plain text	P	0.00
DocumentExtraction status is honest before loading	P	0.00
DocumentExtraction unsupported type and missing file	P	0.00
OCRLanguages capability check does not import the paddle runtime	P	0.00
OCRLanguages codes match what paddleocr accepts	P	0.00
OCRLanguages extractor reports the language it will use	P	0.00
OCRLanguages resolve accepts names codes and aliases	P	0.00
OCRLanguages the indian languages asked for are present	P	0.00
OCRLanguages unknown language falls back to english	P	0.00
OCRLanguages unsupported indian languages are declared	P	0.00
OCRModelCache a failure is retried when the operator asks again	P	0.00
OCRModelCache a half finished download does not count as present	P	0.00
OCRModelCache a machine that has the models is told so every start up	P	0.00
OCRModelCache an empty machine reports nothing cached	P	0.00
OCRModelCache detection alone is not enough to read a page	P	0.00
OCRModelCache prefetch reports every language and never raises	P	7.88
OCRModelCache switching language does not throw the engine away	P	0.00
OCRModelCache the cache directory follows paddle s own variable	P	0.00
OCRModelCache the listing command runs without touching the network	P	0.00
OCRModelCache verification is remembered per language	P	0.00
IngestionPathsEndToEnd a non english report is carried through not dropped	P	0.04
IngestionPathsEndToEnd a pdf is read through its text layer without ocr	P	0.01
IngestionPathsEndToEnd a text document splits into report sized blocks	P	0.02
IngestionPathsEndToEnd corpus aggregation produces the dashboard numbers	P	0.05
IngestionPathsEndToEnd the csv becomes analysed reports	P	0.05
ExtractedDocumentActions analysing one document analyses only its blocks	P	0.20
ExtractedDocumentActions preview shows the document whose button was pressed	P	0.20
ExtractedDocumentActions remove drops only that document and its blocks	P	0.20
ExtractedDocumentActions the controls sit on the row and are not hidden off the edge	P	0.18
SampleReports every sample named in the readme exists	P	0.00
SampleReports the csv imports with its references	P	0.00
SampleReports the multilingual sample covers the languages claimed	P	0.00
SampleReports the sample corpus exercises every offline trigger	P	0.06
SampleReports the scan has no text layer so ocr has to run	P	0.00

Local LLM and translation (23)

Test	Result	s
OllamaEngine a specific tag is not the same model as another	P	0.00
OllamaEngine an untagged name matches the latag	P	0.00
OllamaEngine analysis returns a structured opinion	P	0.00
OllamaEngine availability and models	P	0.00
OllamaEngine missing model is reported with the pull command	P	0.00
OllamaEngine ready is false when the host is unreachable	P	0.00
OllamaEngine ready is false when the model is missing	P	0.00
OllamaEngine replies wrapped in prose are still parsed	P	0.00
OllamaEngine script detection decides when to translate	P	0.00
OllamaEngine translation returns english	P	0.00
OllamaEngine unparsable reply is an error not a crash	P	0.00
OllamaEngine unreachable host degrades quietly	P	0.00
PipelineWithLLM analyze many accepts translations	P	0.04
PipelineWithLLM llm disagreement reaches the review queue	P	0.01
PipelineWithLLM review priority places disagreements first	P	0.00
PipelineWithLLM translated report keeps the original as evidence	P	0.01
PipelineWithLLM unreachable llm leaves the result untouched	P	0.01
PipelineWithLLM verdict is unchanged when the llm is attached	P	0.03
TranslationWithoutAManualProbe a fresh session translates with no probe first	P	0.17
TranslationWithoutAManualProbe an english corpus never asks the translator anything	P	0.13
TranslationWithoutAManualProbe an unreachable host is named rather than left silent	P	0.13
TranslationWithoutAManualProbe readiness is settled once per run not once per report	P	0.13
TranslationWithoutAManualProbe switching translation off is still honoured	P	0.14

Interface (62)

Test	Result	s
WorkflowAndInterface an english report needs no language ceremony	P	0.01
WorkflowAndInterface an untranslated report says so before a decision	P	0.01
WorkflowAndInterface clearing the bench clears the language bar	P	0.01
WorkflowAndInterface dashboard is its own page	P	0.02
WorkflowAndInterface each entry point wears its own skin the same way	P	0.00
WorkflowAndInterface every page can scroll rather than squeeze	P	0.59
WorkflowAndInterface every skin covers the same colours	P	0.00
WorkflowAndInterface map covers every capability in order	P	0.01
WorkflowAndInterface map statuses are settable	P	0.01
WorkflowAndInterface stage button emits its key	P	0.01
WorkflowAndInterface the bench reviews in english and keeps the original	P	0.01
WorkflowAndInterface the decision buttons are never scrolled out of reach	P	0.01
WorkflowAndInterface the interface carries no pictographs	P	0.00
WorkflowAndInterface the scroll helper sets the policies the theme expects	P	0.00
BuildTwoWindowEndToEnd a closed window does not start an update check	P	0.16
BuildTwoWindowEndToEnd a corpus imported twice stays one corpus	P	0.38
BuildTwoWindowEndToEnd a search that matches nothing says so rather than looking broken	P	0.23
BuildTwoWindowEndToEnd documents are read and queued for analysis	P	0.23

Test	Result	s
BuildTwoWindowEndToEnd every language sample is read and analysed	P	0.28
BuildTwoWindowEndToEnd exporting the corpus writes every row	P	0.30
BuildTwoWindowEndToEnd generating a bulletin writes a traceable document	P	0.31
BuildTwoWindowEndToEnd importing a csv goes through the real import path	P	0.30
BuildTwoWindowEndToEnd reviewing a report moves it out of the queue and into the trail	P	0.29
BuildTwoWindowEndToEnd the audit trail carries the session end to end	P	0.29
BuildTwoWindowEndToEnd the log view shows what the session did	P	0.63
BuildTwoWindowEndToEnd the search box filters reports and hotspots alike	P	0.27
BuildTwoWindowEndToEnd the whole console fills in from one import	P	0.51
BuildTwoWindowEndToEnd the workflow map reports what has actually happened	P	0.28
HotspotsPage a populated page says what it adds up to	P	0.25
HotspotsPage an elided cell keeps its full value in a tooltip	P	0.25
HotspotsPage an empty page explains itself and does not repeat its title	P	0.11
HotspotsPage the table comes back once there are hotspots	P	0.25
HotspotsPage the ten columns fit the page instead of scrolling sideways	P	0.23
InterfaceWidgets charts accept data and empty state	P	0.00
InterfaceWidgets dashboard renders a result	P	0.03
InterfaceWidgets data table formats cells	P	0.00
InterfaceWidgets pages scroll instead of squashing	P	0.01
InterfaceWidgets settings view renders state	P	0.00
Theme every referenced asset exists	P	0.00
Theme scroll controls are styled	P	0.00
BlackAndLimeBuild a long field value cannot widen the detail panel	P	0.45
BlackAndLimeBuild column headings are capitals in this build only	P	0.57
BlackAndLimeBuild each entry point builds its own design	P	0.73
BlackAndLimeBuild it carries every capability of the untouched build	P	0.29
BlackAndLimeBuild nothing inside the rail paints the page over it	P	0.41
BlackAndLimeBuild the page heading sits on a band of its own	P	0.44
BlackAndLimeBuild the rail is black and the page is the charcoal	P	0.37
BlackAndLimeBuild the rail is words in capitals and nothing else	P	0.59
BlackAndLimeBuild the skin actually changes the palette	P	0.30
BlackAndLimeBuild the two builds are told apart in the title bar	P	0.30
BlackAndLimeBuild the white and grey skin is still its design token for token	P	0.00
BuildOneInterface the empty detail panel says it is empty	P	0.01
BuildOneInterface the interface carries no pictographs	P	0.00
BuildOneInterface the window remembers where it was	P	0.14
BuildOneInterface the workspace columns can be resized	P	0.01
BuildOneStillWorks the original window analyses the seed corpus	P	0.36
BuildOneStillWorks the two builds coexist in one process	P	0.00
AnalysisWorker falls back to longest cell without known column	P	0.00
AnalysisWorker missing file raises	P	0.00
AnalysisWorker pipeline runs off the main thread	P	0.04
AnalysisWorker reads named description column with reference	P	0.02
ImportSpeed the console adds little to the cost of analysis	P	0.58

Audit and logging (18)

Test	Result	s
AuditTrail a corrupt line costs only that line	P	0.00
AuditTrail an entry names who did it	P	0.00
AuditTrail an unwritable location is reported not raised	P	0.00
AuditTrail both kinds are recorded and told apart	P	0.00
AuditTrail export can be narrowed to one kind	P	0.00
AuditTrail export writes oldest first for a reader	P	0.00
AuditTrail it is append only not rewritten	P	0.00
AuditTrail the newest entry is the first one read back	P	0.00
AuditTrail the trail survives a restart	P	0.00
SystemLogging listener receives records	P	0.00
SystemLogging ring buffer captures and filters	P	0.00
TamperEvidentTrail a deleted entry is caught	P	0.00
TamperEvidentTrail an edited entry is caught where it was edited	P	0.00
TamperEvidentTrail an inserted entry is caught	P	0.00
TamperEvidentTrail an untouched trail verifies	P	0.00
TamperEvidentTrail entries from before chaining are accepted as history	P	0.00
TamperEvidentTrail every entry names the signed in account	P	0.00
TamperEvidentTrail two holders of the file keep one chain	P	0.01

Sign-in and roles (25)

Test	Result	s
Accounts a disabled account cannot sign in	P	0.01
Accounts a lockout ends by itself	P	0.00
Accounts a one time password works once then must be replaced	P	0.01
Accounts a right password signs in and names the person	P	0.00
Accounts a wrong password and an unknown user read the same	P	0.00
Accounts an unreadable file is an error not an empty store	P	0.00
Accounts each role can do everything the one before can	P	0.00
Accounts five wrong passwords lock the account	P	0.01
Accounts saving leaves no half written file	P	0.00
Accounts the last administrator cannot be removed	P	0.00
Accounts the password itself is never written	P	0.00
Accounts two accounts with one password store different digests	P	0.00
Accounts what is refused at creation	P	0.00
SignInAndAttribution a decision is recorded under the signed in person only	P	3.00
SignInAndAttribution a guarded action still accepts a signal argument	P	0.17
SignInAndAttribution a one time password leads to choosing a real one	P	0.01
SignInAndAttribution a refused sign in is recorded with the name that was tried	P	0.01
SignInAndAttribution a viewer is refused and the refusal is recorded	P	0.12
SignInAndAttribution an analyst can analyse but not decide or clear	P	0.17
SignInAndAttribution every analysed report names who analysed it	P	0.15
SignInAndAttribution first run creates the administrator and signs them in	P	0.01
SignInAndAttribution only an administrator manages accounts	P	0.29
SignInAndAttribution signing out is recorded and hands back to sign in	P	0.16

Test	Result	s
SignInAndAttribution the activity page tallies each person	P	0.20
SignInAndAttribution the entry point will not open without a sign in	P	0.00

Release and install (28)

Test	Result	s
ReleaseWorkflow pyinstaller spec is present	P	0.00
ReleaseWorkflow the installer falls back to the packages own version	P	0.00
ReleaseWorkflow the spec builds the shipped interface	P	0.00
ReleaseWorkflow workflow exists and triggers on version tags	P	0.00
ReleaseWorkflow workflow publishes what the updater looks for	P	0.00
ReleaseWorkflow workflow stamps the version from the tag	P	0.00
UpdateChecker a packaged build is offered the installer	P	0.03
UpdateChecker download verifies the checksum	P	0.09
UpdateChecker finds a newer release	P	0.03
UpdateChecker no asset for this platform is reported not raised	P	0.00
UpdateChecker picks the asset for this platform	P	0.03
UpdateChecker prereleases are ignored unless asked for	P	0.09
UpdateChecker release without checksums is refused	P	0.06
UpdateChecker repository without a release is not a failure	P	0.07
UpdateChecker same or older release is not an update	P	0.07
UpdateChecker source checkout is told to pull not to install	P	0.03
UpdateChecker tampered download is refused and deleted	P	0.09
UpdateChecker unreachable api degrades quietly	P	0.03
UpdateChecker unreachable repository says which repository	P	0.06
Versioning garbage never triggers an update	P	0.00
Versioning ordering	P	0.00
Versioning parse and normalise	P	0.00
Versioning prereleases sort below their release	P	0.00
Versioning the package reports one version	P	0.00
Versioning version line is machine editable	P	0.00
WhereAnInstalledBuildWrites a checkout writes beside the code	P	0.00
WhereAnInstalledBuildWrites a frozen build writes under the user data directory	P	0.00
WhereAnInstalledBuildWrites an explicit path still wins	P	0.00

Appendix B How the numbers were produced

Figure	Produced by
Test results	python -m unittest discover -p "test_*.py" with QT_QPA_PLATFORM=offscreen and SIF_ENCODER=hashing; a result class recorded each test's outcome and time.
Engine accuracy	python evaluation/evaluate.py, and the same scoring applied to samples/training_corpus.csv.
Model metrics	MLOpsService.train on samples/training_corpus.csv with its labels, stratified cross-validation.
Speed	SIFPipeline.analyze over the 18 reports of samples/near_miss_reports.csv, offline encoder, after one warm-up call.
Screens	Rendered from the running application, offscreen, against the sample reports.
Code size	16084 lines of application code across sif/, ui/, ui2/ and the entry points; 4581 lines of tests.